

the transition.

3. Set the gradient as either a `fillStyle` or a `strokeStyle` on the context.

It is perhaps easier to think of gradients as a smooth change of color that moves along a line. For example, if you supply points A and B as the arguments to the creation of a gradient, the color will be transitioned for any stroke or fill that moves in the direction of point A to point B.

The canvas drawing context supports two types of gradients:

1. `createLinearGradient(xO, yO, x1, y1)` paints along a line from (xO, yO) to (x1, y1).
2. `createRadialGradient(xO, yO, rO, x1, y1, r1)` paints along a cone between two circles. The first three parameters represent the start circle, with origin (xO, yO) and radius rO. The last three parameters represent the end circle, with origin (x1, y1) and radius r1.

Let us make a linear gradient

```
• <canvas id="d" width="300" height="225"></canvas>
• var d_canvas = document.getElementById("d");
• var context = d_canvas.getContext("2d");
• var my_gradient = context.createLinearGradient(0, 0, 300,
0);
• my_gradient.addColorStop(0, "black");
• my_gradient.addColorStop(1, "white");
• context.fillStyle = my_gradient;
• context.fillRect(0, 0, 300, 225);
```

Let us demystify this code block:-

- ▶ First of all we create an empty canvas of the size 300 X 225.
- ▶ Next we find the `<canvas>` element and access its drawing context
- ▶ Now we can start making the gradient. We make the gradient 300 pixels wide, like our canvas. Because the y values (the 2nd and 4th parameters) are both 0, this gradient will shade evenly from left to right.
- ▶ Once we have a gradient object, we can define the gradient's colors. A gradient has two or more color stops, which can be anywhere along the gradient. To add a color stop, you need to specify its position along the gradient and it can take any value between 0 to 1. Our gradient shades from black to white
- ▶ To render the gradient, you set your `fillStyle` to the gradient and draw a shape, like a rectangle or a line.

Presto! We have our gradient.